

HMC and RStan Replication

STAT 482 – Bayesian Statistics | Spring 2026

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2026-01-01

Paper in One Sentence

[TODO: Summarise the paper's contribution in one sentence for a non-specialist audience.]

This presentation covers:

1. Problem and motivation
2. The Bayesian model
3. Computational method
4. Key findings
5. My R/RStan replication
6. Lessons learned

Paper at a Glance

- ▶ **Authors:** [TODO]
- ▶ **Journal:** [TODO]
- ▶ **Year:** [TODO]
- ▶ **Method:** [TODO: e.g. RStan + HMC]
- ▶ **Domain:** [TODO: e.g. Clinical / Ecology / Economics]
- ▶ **Key tool:** R / RStan

Step 1 – Problem and Motivation

The Research Question

[TODO: State the research question in one or two sentences. What problem does the paper solve? Who cares about the answer?]

Data and Setting:

- ▶ [TODO: dataset description]
- ▶ [TODO: sample size, outcomes measured]
- ▶ [TODO: study design – RCT, observational, etc.]

Why Bayesian?

- ▶ [TODO: reason 1]
- ▶ [TODO: reason 2]
- ▶ [TODO: reason 3]

Where classical methods fall short

[TODO: 1–2 sentences on why a frequentist approach is insufficient or less informative for this specific problem.]

Step 2 – The Bayesian Model

Likelihood

$$[TODO : \quad p(\mathbf{y} \mid \theta) = \dots]$$

[TODO: What are $\mathbf{y}$ (observed data) and θ (parameters) in this paper's context?]

Prior Distribution(s)

$$[TODO : \quad \theta_1 \sim \mathcal{N}(\dots, \dots) \quad [TODO : \quad \theta_2 \sim \dots]$$

Justification: [TODO: one sentence on why these priors are appropriate.]

Posterior – Bayes' Theorem

$$\pi(\theta \mid \mathbf{y}) \propto \underbrace{p(\mathbf{y} \mid \theta)} \times \underbrace{\pi(\theta)}$$

Step 2 (continued) – Model Structure

[TODO: Name of model component / level]

Level 1 – [TODO: label]

$$[TODO : \hat{\theta}_k \mid \theta_k \sim \mathcal{N}(\theta_k, \sigma_k^2)]$$

Level 2 – [TODO: label]

$$[TODO : \theta_k \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2)]$$

Level 3 – [TODO: label]

$$[TODO : \mu \sim \dots, \quad \tau \sim \dots]$$

Key Insight

[TODO: What does this model structure achieve that a simpler model could not? e.g. partial pooling, prior sensitivity, HMC efficiency.]

Parameter interpretation:

- ▶ $\mu = [\text{TODO}]$
- ▶ $\tau = [\text{TODO}]$
- ▶ [TODO: other parameters]

RStan setup:

- ▶ Algorithm: [TODO: HMC/NUTS or other]
- ▶ Chains: [TODO] | Iterations: [TODO]
- ▶ Warm-up: [TODO]
- ▶ adapt_delta: [TODO]
- ▶ Parameterisation: [TODO: centred / non-centred]

Convergence checks:

- ▶ $\hat{R}$: [TODO: value; threshold < 1.01]
- ▶ ESS: [TODO: adequate?]
- ▶ Divergent transitions: [TODO: any?]
- ▶ Trace plots: [TODO: mixed well?]

Trace Plot – [TODO: parameter name]

[Insert your trace plot here]

Good mixing: chains overlap and show no long drifts or stuck regions.

Step 4 – Key Findings

Main Result

[TODO: State the most important finding with a number. Example: “The posterior mean treatment effect is -14.2 min (95% CrI: -22.1 , -6.3), indicating a likely reduction in PACU discharge time.”]

Posterior summaries:

Parameter	Post. mean	95% CrI
[TODO: name]	val	(lo, hi)
[TODO: name]	val	(lo, hi)
[TODO: name]	val	(lo, hi)

Posterior Probability Statement

$$P(\text{[TODO: effect]} > \text{threshold} \mid \mathbf{y}) = \text{[TODO]}$$

In plain language:

[TODO: interpret this for a non-statistician]

[TODO: One further finding – robustness, heterogeneity, or comparison to a frequentist estimate.]

Step 5 – My R / RStan Replication

What I reproduced:

- ▶ [TODO: model name]
- ▶ [TODO: real or simulated data?]
- ▶ [TODO: key result reproduced]

Stan model{} block:

```
// [TODO: paste your Stan model block]
// Show the parameters{} and model{} sections
parameters {
  real beta1;      // treatment effect
  real<lower=0> sigma;
}
model {
  beta1 ~ normal(0, 10); // prior
  y ~ normal(beta0 + beta1 * x, sigma);
}
```

My Key Output

[Insert your main plot or table here]
e.g. posterior density, forest plot, credible interval plot, shrinkage plot

Matches the paper?

[TODO: Qualitative comparison – direction, magnitude, and uncertainty.]

Three things I understand better now:

1. [TODO: Statistical concept]

- [TODO: What you understand now that you did not before reading the paper – be specific.]

2. [TODO: Link to STAT 482]

- [TODO: Connect to a lecture topic – prior choice, MCMC diagnostics, hierarchical models, ...]

3. [TODO: Practical takeaway]

- [TODO: One thing you will do differently in your own Bayesian analyses.]

Open Question

[TODO: One aspect the paper does not address that you think would strengthen or extend the analysis.]

Connection to This Course

[TODO: Name the STAT 482 lecture or topic this paper most directly illustrates or extends.]

Assigned Paper

[TODO: Author(s), Journal, Year]

DOI: [TODO: insert DOI]

R Packages Used in My Replication

- ▶ `rstan` – Bayesian model fitting via Stan
- ▶ `ggmcmc` – MCMC diagnostic plots
- ▶ `HDInterval` – Highest density credible intervals
- ▶ [TODO: any additional packages used]

Thank you – Questions Welcome

R/Stan code available upon request.